



Lecture Notes

Institute Name:	Silver Oak College of Computer Application		
Programme Name:	MCA/MSCIT/ MSC DS	Semester:	1
Course Code:	MDC5001C	Course Name:	Foundation of Mathematics
Unit No.:	1	Unit Name:	Introduction to Combinatorics and Its Applications

Unit 1: Introduction to Combinatorics and Its Applications

Combinatorics is the branch of mathematics concerned with counting, arrangement, and selection of objects. It provides the foundational tools used throughout probability theory, statistics, computer science, and data science whenever we need to answer the question, “In how many ways can this happen?” This unit introduces the fundamental counting principles, permutations and combinations, the pigeonhole principle, the binomial theorem and Pascal’s triangle, the inclusion-exclusion principle, derangements, multisets, integer partitions, and recurrence relations.

Learning Objectives

- State and apply the addition and multiplication rules of counting.
- Distinguish between permutations and combinations, and compute each with and without repetition.
- Apply the pigeonhole principle to prove the existence of repeated outcomes.
- Use the binomial theorem and Pascal’s triangle to expand expressions and compute coefficients.
- Apply the inclusion-exclusion principle to problems involving two or more overlapping sets.
- Compute derangements and apply them to real-world arrangement problems.



- Work with multisets, integer partitions, and their diagrammatic representations.
- Set up and interpret simple recurrence relations.

1. The Fundamental Counting Principle

If one event can occur in m ways and a second, independent event can occur in n ways, then the two events together can occur in $m \times n$ ways. This simple idea underlies all of counting theory: it lets us systematically determine the total number of possible outcomes when several independent choices are combined in sequence.

Example. Consider getting dressed in the morning with 3 shirt choices and 4 pant choices. Since the choice of shirt is independent of the choice of pants, the total number of outfits is $3 \times 4 = 12$ unique outfits.

Addition Rule — Mutually Exclusive Events

The addition rule applies when events cannot happen simultaneously — you must choose one option OR another, never both. If event A can occur in m ways and event B (mutually exclusive of A) can occur in n ways, then A or B can occur in $m + n$ ways.

Example. At a restaurant, a diner may choose from 5 cake options or 3 ice-cream options, but not both in this scenario. The total number of dessert choices is $5 + 3 = 8$.

Multiplication Rule — Compound Choices

The multiplication rule applies to a sequence of independent choices: each step multiplies the possibilities available from the previous step, producing rapid (exponential) growth in the total number of outcomes.

Example. A vehicle license plate consists of 3 letters followed by 2 digits. The number of possible plates is $26^3 \times 10^2 = 1,757,600$.

2. Permutations — Order Matters

A permutation is an arrangement of objects in a specific order. Changing the order of the same objects produces a different permutation. The number of ways to arrange n distinct objects is $n!$ (n factorial), and the number of ways to arrange r objects chosen from n distinct objects is given by:

$$P(n, r) = n! / (n - r)!$$

Example. Three runners finish a race in 1st, 2nd and 3rd place. The number of possible finishing orders is $3! = 3 \times 2 \times 1 = 6$.

Permutations with Repetition Allowed

When repetition is allowed, every position has the full set of choices available regardless of what was chosen in earlier positions. The number of permutations of length r drawn (with repetition) from n objects is n^r .

Example. A 4-digit PIN is formed from the digits 0–9 with repetition allowed. Each of the 4 positions has 10 possible digits, giving $10^4 = 10,000$ possible PINs.

3. Combinations — Order Does Not Matter

A combination counts the number of ways to select objects without regard to order; rearranging the same selected items does not create a new combination. The number of ways to choose r objects from n distinct objects is:

$$C(n, r) = n! / [r! (n - r)!]$$

Example. A team of 3 members is chosen from a group of 10 people. The number of possible teams is $C(10, 3) = 120$. Note that the team {Alice, Bob, Charlie} is the same as {Charlie, Alice, Bob} — order does not matter in combinations.

4. Key Identities and Relationships

- **Permutations & Combinations:** Permutations equal combinations multiplied by the number of ways to arrange the selected items: $P(n, r) = C(n, r) \times r!$
- **Combined Principles:** The addition and multiplication rules work together to solve complex, multi-step counting problems that involve both choices and sequences.
- **Factorial Growth:** Factorials grow extremely quickly: $5! = 120$; $10! = 3,628,800$; $20! \approx 2.43 \times 10^{18}$.

5. Real-World Applications

- **Security & Cryptography:** Password and code generation use permutations with repetition to build secure authentication systems.
- **Lottery & Probability:** Combinations are used to calculate odds in lotteries and games of chance, showing why winning is so unlikely.
- **Scheduling & Arrangements:** Seating charts, meeting schedules and event planning rely on permutations without repetition.
- **Decision Analysis:** Probability trees built from permutations and combinations support decision analysis in business, statistics and risk assessment.

6. Worked Examples — Permutations and Combinations

Example 1. Find the number of words, with or without meaning, that can be formed using the letters of the word “PARK”.

Solution. The word PARK has 4 distinct letters. The number of arrangements is $4! = 4 \times 3 \times 2 \times 1 = 24$ words.

Example 2. How many 6-digit codes can be generated using the digits 0–9 (a) if repetition is allowed, and (b) if repetition is not allowed?

Solution. (a) With repetition: $10^6 = 1,000,000$ codes. (b) Without repetition: $10 \times 9 \times 8 \times 7 \times 6 \times 5 = 151,200$ codes.

Example 3. Find the number of different words that can be formed from the letters of the word “TREAT” so that the vowels (E, A) are always together.

Solution. Treat the two vowels (E, A) as a single block. The word then effectively has 4 units to arrange: {EA}, T, R, T, of which T repeats twice, giving $4!/2!$ arrangements of the units, multiplied by $2!$ ways to arrange the vowels within their block.

Example 4. The word EDUCATION has 9 distinct letters. How many 3-letter permutations (words) can be formed using its letters?

Solution. The number of 3-letter permutations is $P(9, 3) = 9!/6! = 9 \times 8 \times 7 = 504$.

Example 5. Out of a group of 5 people, a pair needs to be formed. In how many ways can this be done?

Solution. This is a combination problem: $C(5, 2) = 5!/(2!3!) = 10$ ways.

Example 6. Find the number of 4-letter combinations that can be made from the letters of the word DRIVEN.

Solution. DRIVEN has 6 distinct letters. The number of 4-letter combinations is $C(6, 4) = 6!/(4!2!) = 15$.

7. The Pigeonhole Principle

If more pigeons are placed into pigeonholes than there are pigeonholes available, then at least one pigeonhole must contain two or more pigeons. This seemingly obvious statement yields surprisingly powerful results and is widely used to prove the existence of an outcome without constructing it explicitly.

Classic Example. Among any group of 13 people, at least two people must share the same birth month, since there are only 12 months available and 13 people must be distributed among them.

Example. In a crowd of 367 people, at least two people will share the same birthday, because there are only 366 possible birthdays (including 29 February).

□ Applications:

- Proving the existence of duplicates without constructing them explicitly.
- Guaranteeing repeats in sequences and data structures.
- Solving logical puzzles and competition-level problems.
- Establishing results in number theory and theoretical computer science.

Example — Class Scheduling. A college can schedule classes in 35 different time periods. If there are 679 different classes, what is the minimum number of rooms needed?

Solution. By the pigeonhole principle, distributing 679 classes across 35 time periods requires at least $\lceil 679/35 \rceil = 20$ rooms running in parallel during the busiest period.

Example — Pairing. Select any 7 numbers from 1 to 11. Show that two of them must sum to 12.

Solution. Group the numbers 1–11 into 6 “holes” by pairs that sum to 12: (1,11), (2,10), (3,9), (4,8), (5,7), and 6 alone. Choosing any 7 numbers means at least two must fall into the same pair, guaranteeing two numbers that sum to 12.

8. Pascal's Triangle and the Binomial Theorem

Pascal's triangle is a triangular array of numbers in which each entry is the sum of the two entries directly above it, and the entries in row n are precisely the binomial coefficients $C(n, 0), C(n, 1), \dots, C(n, n)$. The binomial theorem expresses the expansion of $(x + y)^n$ in terms of these coefficients:

$$(x + y)^n = \sum C(n, k) \cdot x^{n-k} \cdot y^k, \text{ for } k = 0 \text{ to } n$$

- **Pascal's Triangle:** Provides a quick reference for binomial coefficients.
- Reveals hidden patterns, such as the Fibonacci sequence appearing along its diagonals.
- Generates combinatorial identities and connects to fractal geometry.
- **Binomial Theorem:** Calculates probabilities in binomial distributions.
- Expands polynomials efficiently and approximates functions in numerical methods.
- **Pigeonhole Principle (recap):** Guarantees existence of duplicates, proves number-theoretic results, and solves competition-level puzzles.

Example 1. Expand $(1 - 2x)^5$.

Solution. $(1 - 2x)^5 = 1 - 10x + 40x^2 - 80x^3 + 80x^4 - 32x^5$.

Example 2. Using the binomial theorem, find 96^3 .

Solution. $96^3 = (100 - 4)^3 = C(3,0)(100)^3 - C(3,1)(100)^2(4) + C(3,2)(100)(4)^2 - C(3,3)(4)^3 = 1,000,000 - 120,000 + 4,800 - 64 = 884,736$.

Example 3. Using the binomial theorem, find 102^5 .

Solution. $102^5 = (100 + 2)^5 = C(5,0)(100)^5 + C(5,1)(100)^4(2) + C(5,2)(100)^3(2)^2 + C(5,3)(100)^2(2)^3 + C(5,4)(100)(2)^4 + C(5,5)(2)^5 = 11,040,808,032$.

9. The Inclusion-Exclusion Principle

The Principle of Inclusion and Exclusion (PIE) is a counting technique used to find the size of the union of several overlapping sets by systematically adding and subtracting the sizes of their intersections, so that no element is counted more than once.

For two sets A and B , the size of their union is:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

For three sets A , B and C , the size of their union is:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Example 1. In a class of 100 students, 60 study Math and 45 study Science, with 20 studying both. How many students study Math or Science?

Solution. $|A \cup B| = 60 + 45 - 20 = 85$ students.

Example 2. In a survey of 120 people, 80 like tea, 70 like coffee, and 50 like both. How many like tea or coffee?

Solution. $|A \cup B| = 80 + 70 - 50 = 100$ people.

Example 3. In a survey of 200 people: 120 like pizza, 100 like burgers, 80 like tacos; 60 like both pizza and burgers, 40 like both burgers and tacos, 30 like both pizza and tacos, and 20 like all three. How many people like at least one of the three foods?

Solution. $|A \cup B \cup C| = 120 + 100 + 80 - 60 - 30 - 40 + 20 = 190$ people.

Example 4. In a university of 300 students: 150 take Math, 120 take Physics, 100 take Chemistry; 80 take both Math and Physics, 60 take both Physics and Chemistry, 50 take both Math and Chemistry, and 30 take all three. How many students take at least one subject?

Solution. $|A \cup B \cup C| = 150 + 120 + 100 - 80 - 60 - 50 + 30 = 210$ students.

Practice Problems

- In a group of 200 people, 120 like chocolate, 90 like vanilla, and 50 like both. How many people like either chocolate or vanilla?
- In a class of 100 students, 70 play football, 60 play basketball, and 50 play cricket; 30 play both football and basketball, 25 play both basketball and cricket, 20 play both football and cricket, and 15 play all three. How many students play at least one sport?
- In a school of 300 students, 180 are enrolled in Math, 150 in Science, and 120 in English; 90 in both Math and Science, 80 in both Science and English, 70 in both Math and English, and 50 in all three. How many students are enrolled in at least one subject?

10. Derangements and Applications

A derangement is a permutation of the elements of a set in which no element appears in its original (natural) position — every element is displaced. The number of derangements of n objects is denoted D_n or $D(n)$.

Illustration. Suppose there are three balls and three boxes of green, blue and red colours, and we want to place the balls into the boxes so that no ball is placed in the box of its own colour. This is an example of a derangement; it can be done in exactly two ways.

Building Up the Derangement Formula

Case 1 ($n = 1$). Consider one letter L_1 and its envelope E_1 . There is no way to place the single letter into the wrong envelope, so $D(1) = 0$.

Case 2 ($n = 2$). Consider two letters L_1, L_2 and their envelopes E_1, E_2 . Only one arrangement puts both letters in the wrong envelope (L_1 into E_2 and L_2 into E_1), so $D(2) = 1$.

Case 3 ($n = 3$). Consider three letters L_1, L_2, L_3 and their envelopes E_1, E_2, E_3 . There are 2 ways to place every letter into a wrong envelope (for example, L_1 can go into E_2 or E_3), so $D(3) = 2$.

Following the same reasoning, $D(4) = 9$, $D(5) = 44$, $D(6) = 265$, and so on. In general, the number of derangements of n objects is given by:

$$D(n) = n! \times (1 - 1/1! + 1/2! - 1/3! + \dots + (-1)^n/n!)$$

Example 1. In how many ways can 7 letters be placed into 7 envelopes such that exactly 3 go into the correct envelope?

Solution. The 3 correctly placed envelopes can be chosen in $C(7,3) = 35$ ways. The remaining 4 letters must all be deranged: $D(4) = 9$. Total arrangements = $35 \times 9 = 315$.

Example 2. In how many ways can a dancing couple be formed from 3 boys and 3 girls so that no boy dances with his own girlfriend?

Solution. This is a derangement of 3 items: $D(3) = 3!(1 - 1/1! + 1/2! - 1/3!) = 6 \times (2/6) = 2$ ways.

11. Multisets

A multiset is a collection of objects, similar to a set, but which allows an object to appear more than once (the order of the elements still does not matter). For example, $\{1, 1, 2, 5, 5, 7\}$ is a multiset of size 6.

- **Examples:** $\{1, 1, 2\}$ — the element 1 is repeated.
- $\{a, a, a, b, b\}$ — three occurrences of a and two of b .
- $\{\text{red, red, blue, blue, blue}\}$ — a multiset of colours where “red” appears twice and “blue” appears three times.

Permutations of a Multiset

The number of distinct ordered arrangements (permutations with repetition) of a multiset of n objects, where object kind 1 occurs with multiplicity n_1 , kind 2 with multiplicity n_2 , ..., and kind k with multiplicity n_k , is:

$$n! / (n_1! \times n_2! \times \dots \times n_k!)$$

Example. Consider the multiset $\{A, A, B, B, C\}$ and find all permutations of this multiset.

Solution. The total number of elements is $n = 5$, with A appearing twice, B appearing twice, and C appearing once. The number of distinct permutations is $P = 5! / (2! \times 2! \times 1!) = 120 / 4 = 30$. So there are 30 distinct permutations of the multiset $\{A, A, B, B, C\}$.

Combinations of a Multiset

When choosing r elements from a multiset with repetition allowed among k distinct kinds, the number of combinations is $C(r + k - 1, r)$. For example, choosing 3 elements (with repetition allowed) from the multiset $\{A, B, C\}$ gives $C(3 + 3 - 1, 3) = C(5, 3) = 10$ possible combinations.

12. Integer Partitions and Compositions

An integer partition of a positive integer n is a way of writing n as a sum of positive integers, where the order of the summands does not matter. For example, the integer 4 can be partitioned in five distinct ways:

$$\begin{aligned} &4 \\ &3 + 1 \\ &2 + 2 \\ &2 + 1 + 1 \\ &1 + 1 + 1 + 1 \end{aligned}$$

Similarly, the integer 5 has seven partitions: 5; 4 + 1; 3 + 2; 3 + 1 + 1; 2 + 2 + 1; 2 + 1 + 1 + 1; and 1 + 1 + 1 + 1 + 1. A partition such as 2 + 2 + 1 may also be written as the tuple (2, 2, 1), or more compactly as $(2^2, 1)$, where the superscript denotes the number of times a part is repeated.

Diagrammatic Representations of Partitions

Ferrers Diagram. A Ferrers diagram represents a partition using rows of dots, one row per part, with each row length equal to the size of that part. For example, the partition 6 + 4 + 3 + 1 of the number 14 is shown as 4 rows of dots containing 6, 4, 3, and 1 dots respectively, lined up on the left.

Young Diagram. A Young diagram represents the same idea using boxes (squares) instead of dots. For instance, the Young diagram for the partition $5 + 4 + 1$ consists of three left-aligned rows of 5, 4, and 1 boxes.

Conjugate and Self-Conjugate Partitions. If the diagram of a partition is reflected along its main diagonal, another partition of the same number is obtained, called its conjugate. A partition that is its own conjugate is called self-conjugate. It can be shown that the number of self-conjugate partitions of n equals the number of partitions of n into distinct odd parts — every odd part can be “folded” at its middle to build a self-conjugate diagram, giving a one-to-one correspondence (bijection) between the two families of partitions.

13. Recurrence Relations and Their Solutions

A recurrence relation is a mathematical expression that defines each term of a sequence in terms of one or more of its preceding terms, together with one or more initial (base) values. Recurrence relations are especially useful for describing quantities that are built up step by step, such as the running time of a recursive algorithm.

Example — Fibonacci Sequence. The Fibonacci sequence is defined by the recurrence $F(n) = F(n - 1) + F(n - 2)$, with base cases $F(0) = 0$ and $F(1) = 1$. This produces the familiar sequence 0, 1, 1, 2, 3, 5, 8, 13, ...

Common Examples of Recurrence Relations

- **Factorial of a number n :** $F(n) = n \times F(n - 1)$, with $F(0) = 1$.
- **Merge Sort (divide-and-conquer sorting):** $T(n) = 2 \cdot T(n/2) + O(n)$.
- **Tower of Hanoi:** $H(n) = 2 \cdot H(n - 1) + 1$, with $H(1) = 1$.
- **Binary Search:** $T(n) = T(n/2) + 1$.

Recurrence relations of this kind are used to generate divide-and-conquer algorithms, analyse the running time of recursive algorithms, and define the states and transitions used in dynamic programming.

14. Summary and Key Takeaways

- **Addition & Multiplication Rules:** Use the addition rule for mutually exclusive events (choices that cannot occur together) and the multiplication rule for sequential, independent events (choices made in succession).
- **Permutations:** Count ordered arrangements, where the sequence in which objects are placed matters significantly.
- **Combinations:** Count unordered selections, where the arrangement of the chosen items is irrelevant.
- **Pigeonhole Principle:** Guarantees the existence of a repeated outcome whenever there are more items than categories.
- **Binomial Theorem:** Expand powers of binomials and connect to combinatorial coefficients through Pascal's triangle.
- **Inclusion-Exclusion:** Compute the size of a union of overlapping sets without double-counting shared elements.
- **Derangements:** Count arrangements in which no item occupies its original position.
- **Multisets & Partitions:** Extend counting to collections with repeated elements, and to the ways a whole number can be broken into parts.
- **Recurrence Relations:** Describe a sequence in terms of its own earlier terms, forming the basis of recursive algorithm analysis.

Mastering these fundamental principles equips students with powerful problem-solving tools for probability, statistics and combinatorics — skills that are essential in data science, computer science and advanced mathematics.